Analysis of Agricultural Loans Granted by Banks in Paraguay

Pilar Ruiz Diaz — Mathias Chaparro — Tanya godoy

Contents

List of Figures

List of Tables

Analysis of Agricultural Loans Granted by Banks in Paraguay

***Abstract*—This study explores the dynamics of agricultural credit allocation by banking institutions in Paraguay, aiming to assess how financial support is distributed across the sector and how effectively it meets the needs of producers. By analyzing loan data using time series and the statistical modeling techniques we've been studying, the research identifies key trends, seasonal behaviors, and risk patterns that influence credit performance. Special attention is given to the accessibility of credit for small and medium-sized producers. The findings provide insights into structural inefficiencies in the current financial system and offer evidence-based recommendations to enhance equitable access to agricultural financing and improve the resilience of rural economies.**

I. Introduction

Agriculture is a key driver of Paraguay's economy, yet many producers—especially small and medium-sized ones—struggle to access adequate financing. While banks play a central role in supporting agricultural activity, questions remain about how effectively credit is allocated and whether it meets the sector's diverse needs.

This study analyzes historical agricultural loan data from Paraguayan banks using time series and statistical methods. The goal is to identify patterns, trends, and inefficiencies in credit distribution, offering insights to improve financial inclusion and support sustainable agricultural development.

II. Problem Statement

The agricultural sector in Paraguay contributes around 20% of GDP and employs about 24% of the active population. However, access to financing remains a challenge, particularly for small and medium-sized producers. As of September 2024, agricultural loans represented 17% of total bank credit, but concerns remain about loan concentration, repayment risks, and limited access to structured debt solutions.

While commercial banks are key players, the *Crédito Agrícola de Habilitación (CAH)* plays a vital role in expanding rural credit access. In 2024, CAH served over 71,000 clients, disbursing more than G. 495 billion, with 40% of its clients being women and 27% youth. New products like *Inversión Productiva 2.0*, supported by FOGAPY, offer long-term financing with interest rates around 11.5%.

Despite this, there is a lack of public studies that analyze aggregated agricultural credit data using time series tools. Incorporating CAH's role and credit dynamics into such analysis would provide a clearer picture of financing trends and support better-informed policy decisions for the sector.

III. Methodology and Steps

This section details the step-by-step methodology followed for the analysis of agricultural loan data, from initial setup to advanced time series modeling.

A. Importation of Libraries

The analysis began by importing essential Python libraries to handle data manipulation, visualization, statistical analysis, and time series modeling:

- `pandas` for data structuring and operations.
- `numpy` for numerical computations.
- `warnings` to manage system warnings during execution.
- `matplotlib.pyplot` and `seaborn` for data visualization.
- `statsmodels.tsa.seasonal.seasonal_decompose` for time series decomposition.
- `statsmodels.graphics.tsaplots.plot_acf`, `plot_pacf` for autocorrelation plots.
- `statsmodels.api as sm` for statistical modeling, including OLS, ARIMA, and SARIMA.
- `sklearn.metrics.mean_squared_error` for model evaluation.
- `statsmodels.tsa.stattools.adfuller` for stationarity testing.
- `scipy.stats` for various statistical tests (e.g., Shapiro-Wilk for normality).

B. Data Loading and Preprocessing

The core data, representing agricultural credit per sector, was loaded and transformed:

- Data Loading: The script first attempts to load data from a CSV file (e.g., `Boletin_Procesado.xlsx - Cred_por_sector.csv`). If successful, it proceeds; otherwise, it would typically load from an Excel file containing multiple sheets.
- Sector Filtering: Data was filtered to include only the 'AGRICULTURA' sector.
- Date Conversion: The 'Fecha' (Date) column was converted to datetime objects, assuming a 'YYYY/MM' format and appending '/01' to create a valid month-start date.
- Data Reshaping (Melt): The dataset was transformed from a wide format (where each bank was a column) to a long format using `pd.melt()`. This created dedicated 'Banco' (Bank) and 'Credito' (Credit) columns.

- Bank Name Standardization: Bank names were standardized to account for mergers such as the onces that ocurred with SUDAMERIS and GNB, and Vision with UENO Bank.
- Aggregation: Data was grouped by 'Fecha', 'Moneda' (Currency), and 'Banco', summing the 'Credito' values to ensure unique monthly entries per bank and currency.
- Aggregated Dataframe (without Currency): A separate DataFrame, `df_nuevo_banco`, was created by summing credits across currencies, providing total credit by bank and date.
- Data Cleaning: 'Credito' values were converted to numeric, and rows with 'NaN' or zero values in 'Credito' were removed to ensure data integrity for subsequent analyses.

C. Exploratory Data Analysis (EDA) and General Visualizations

Initial exploratory analysis and visualizations were performed to understand the aggregated credit data:

- Monthly Average Credit: A line plot showing the average total credit for each month across all years, revealing general seasonal patterns.
- Monthly Credit per Year Comparison: A multi-line plot displaying monthly credit totals for each year, allowing for visual comparison of seasonal patterns and trends across different years.

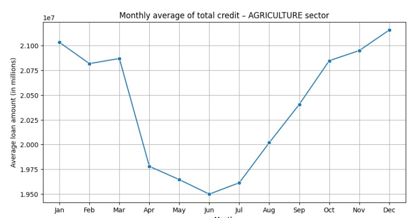

Fig. 1. Agricultural credits per year

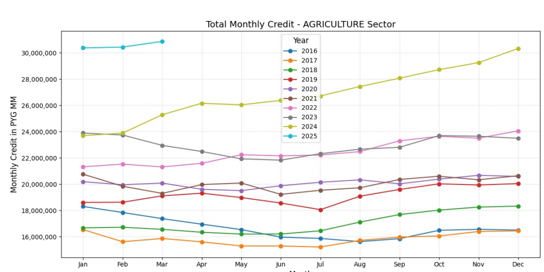

Fig. 2. Agricultural credits per year

D. Time Series Decomposition (Additive and Multiplicative Models)

- Series Aggregation: The total credit was summed by date and set to a monthly start frequency and residuals from both models were plotted and their variances compared to identify which model better captures the underlying patterns (typically, the one with lower residual variance is preferred). The seasonal components of both models were also plotted side-by-side for visual comparison. Then, individual time series plots were generated for each bank:

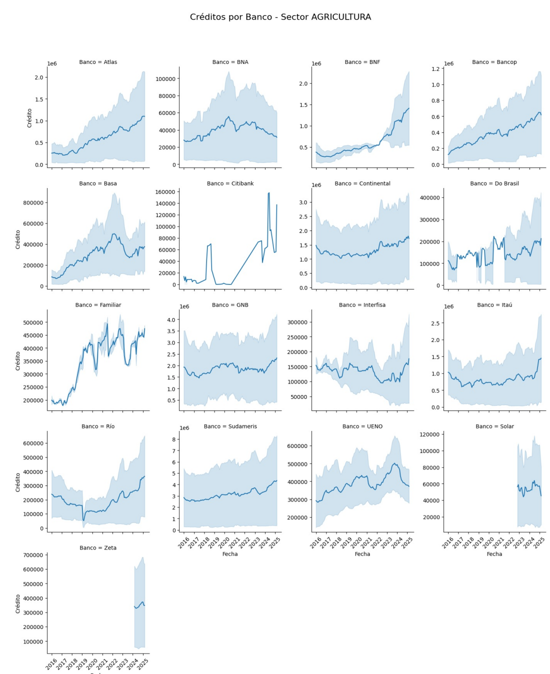

Fig. 3. Agricultural credits per Bank

E. Coefficient of Variation (CV) Analysis and Currency Distribution

The volatility and distribution of credit were analyzed, considering the impact of currency:

- Descriptive Statistics by Currency: Basic descriptive statistics (mean, std, quartiles) for 'Credito' were computed, grouped by 'Moneda' (MN and ME).
- CV Calculation: The Coefficient of Variation (CV = standard deviation / mean) was calculated for each bank, first disaggregated by currency, and then for the total credit across all currencies.
- CV Comparison Plot: A bar plot was generated to compare the CVs of top banks, showing the relative volatility of their credit portfolios with and without distinguishing by currency.
- Credit Distribution Plots: Box plots and histograms were used to visualize the distribution of 'Credito', segmented

by 'Moneda', to highlight differences in credit magnitudes and patterns between National Currency (MN) and Foreign Currency (ME).
- After applying these methods, we noticed, over 85% of the of the loans granted where in Foreign Currency (ME) and the data was more stable, so we stayed with this segment of the dataset.

F. Stationarity Tests (ADF) and Residual Analysis

As previously mentioned, we continued taking the steps to prepare the data for ARIMA/SARIMA modeling by testing for stationarity and analyzing model residuals:

- ADF Test Function: A helper function, `adf_test()`, was defined to perform the Augmented Dickey-Fuller (ADF) test, print its statistics (ADF statistic, p-value, critical values), and indicate if the series is stationary. The ADF test was applied to the original total credit series in ME. As the series was non-stationary, common transformations (logarithm and first-order differencing) were applied to it. (e.g., `np.log()` and `.diff()`), and the ADF test was reapplied to the transformed series (e.g., log-differenced series) to confirm stationarity.
- ACF and PACF Plots: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were generated for the stationary (log-differenced) series. These plots are crucial for identifying the appropriate orders (p, q for ARIMA; P, Q for SARIMA) for the time series models.

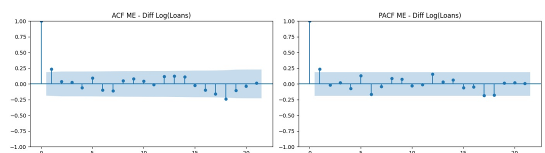

Fig. 4. Agricultural credits per Bank

G. SARIMA and ARIMA Modeling

Finally, Seasonal AutoRegressive Integrated Moving Average (SARIMA) and AutoRegressive Integrated Moving Average (ARIMA) models were applied for forecasting adn comparing. The total credit series in Foreign Currency (ME) was used for this purpose.

Series Preparation: The 'Credito' values for Foreign Currency (ME) were summed by date, set to a monthly start frequency, and any 'NaN' values were dropped. SARIMA Model Application:

- Model Definition: A SARIMA model was defined with specific non-seasonal `order=(p,d,q)` (e.g., (1,1,1)) and seasonal `seasonal_order=(P,D,Q,S)` (e.g., (1,1,0,12) for annual seasonality). `enforce_stationarity` and `enforce_invertibility` were set to `False` to allow the model to handle these aspects internally.
- Forecasting: The model was used first to generate forecasts for the same period of time as the real data we had on the original dataset, to see how accurate the prediction was gonna be, and then for a specified number of future months (`num_forecast_months`), including predicted means and 95% confidence intervals.
- Model Evaluation: Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) were calculated by comparing in-sample predictions with actual values to assess the model's accuracy. Giving as a MAPE result of 3.67%, which is a low value, indicating that we had a good percent of precision of the forecast.
- Visualization: A plot was generated showing the actual credit values, the in-sample fitted values, and the future forecast with its confidence interval.

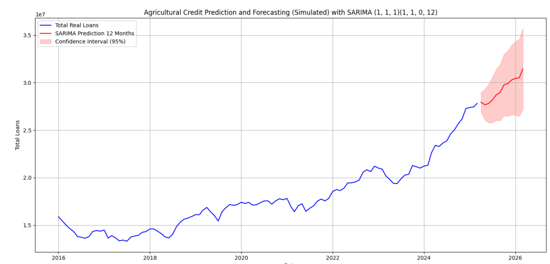

Fig. 5. SARIMA model result

ARIMA Model Application:

- The process was largely repeated for an ARIMA model, which is a non-seasonal version of SARIMA, it was defined with a non-seasonal `order=(p,d,q)` (e.g., (1,1,1)). Similar steps as the SARIMA model were performed to fit the model, display its summary, generate forecasts, calculate RMSE and MAPE, and visualize the results.

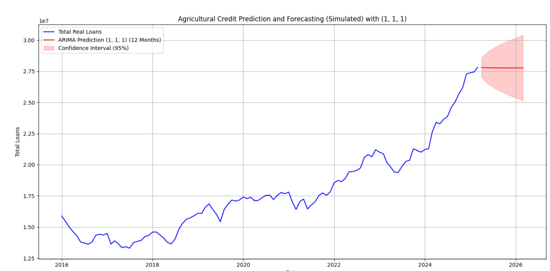

Fig. 6. ARIMA model result

IV. SARIMA AND ARIMA COMPARISION

Based on the statistical information criteria (Log Likelihood, AIC, BIC) and the Ljung-Box test for residual autocorrelation, the SARIMA model appeared to be the better choice overall. Because it captured the underlying patterns of the time series more effectively, especially the seasonality, as indicated by its higher log-likelihood, lower AIC/BIC, and residuals that resemble white noise. However, the ARIMA model shows lower RMSE and MAPE, which are direct measures of forecasting accuracy, suggesting that while SARIMA is a better statistical fit for the overall series structure (including seasonality), the ARIMA model's point predictions might be closer to the actual values on average in some specific instances. The statistical summaries (AIC, BIC, and Ljung-Box) strongly support this conclusion, as SARIMA effectively handles the seasonality that ARIMA misses, leading to cleaner residuals. The lower RMSE/MAPE of ARIMA could be due to specific characteristics of the data or potentially overfitting in a non-seasonal context if the seasonal pattern isn't perfectly regular.

Conclusion: Model Selection for Agricultural Credits

Following a comprehensive analysis and evaluation of both ARIMA and SARIMA models for agricultural credits, the SARIMA(1,1,1)(1,1,0,12) model has been selected as the optimal choice. This decision was based on the following key indicators:

Key Indicator	ARIMA(1,1,1)	SARIMA(1,1,1)(1,1,0,12)	Rationale for Selection
AIC (Akaike Information Criterion)	3.165,36	2.466,73	SARIMA exhibits a significantly lower AIC, indicating a better fit when penalizing model complexity.
BIC (Bayesian Information Criterion)	3.173,46	2.476,50	Similar to AIC, SARIMA's BIC is remarkably lower, suggesting it is a more parsimonious and efficient model.
Ljung-Box (Q) Prob(Q)	0.01 (NOT White Noise)	0.95 (White Noise)	This is the decisive criterion. SARIMA successfully ensures that residuals are white noise, meaning it has effectively captured all autocorrelation in the series. ARIMA fails this fundamental test, indicating the model is misspecified.
Significant Seasonal Term	N/A	ar.S.L12 (p=0.000)	SARIMA identifies and models a strong seasonal dependency at lag 12, which is consistent with the agricultural nature of the data.
Heteroskedasticity (Prob(H))	0,01	0	Both models show signs of heteroskedasticity, suggesting that the variance of the errors is not constant. This is an area for future investigation (e.g., GARCH models) to improve the reliability of confidence intervals.
MAPE (Mean Absolute Percentage Error)	2,72%	3,67%	Although ARIMA's MAPE was slightly lower in the evaluated forecast window, SARIMA's statistical superiority (especially in residual whitening) takes precedence for model validity. Both MAPEs are very low, indicating good accuracy.
Estimation Warnings	Singular Co. Matrix	Singular Co. Matrix	Both models present warnings regarding the covariance matrix, indicating that the standard errors of the coefficients may be unstable. This must be considered when interpreting the individual significance of coefficients.

TABLE I. SARIMA AND ARIMA MODELS RESULT COMPARISION

V. Research Methodology

The research methodology involved a sequential process of data-driven analysis using Python programming. Data collection was based on pre-existing CSV and Excel files containing historical agricultural loan data we got from the BCP web page. The methodology was applied in a systematic manner to understand the underlying structure of the data and to build predictive models. Each analytical step, from data cleaning to model fitting, was executed programmatically, allowing for reproducibility and consistency. The iterative process of testing for stationarity and applying transformations was central to preparing the time series for robust modeling. The analytical techniques were selected based on their established efficacy in time series forecasting and statistical inference within financial contexts.

VI. Analysis and Interpretation

The analysis and interpretation phase focused on extracting meaningful insights from each step of the data processing and modeling.

- **Data Preprocessing:** The initial analysis revealed the importance of careful data cleaning and reshaping, highlighting issues like inconsistent bank names and the need for date standardization to enable time series analysis.
- **Exploratory Data Analysis (EDA):** EDA provided preliminary insights into the distribution of credit across banks and identified overall annual and monthly trends. Visualizations suggested inherent seasonality and underlying trends in the credit allocation, prompting further investigation.
- **Time Series Decomposition:** This step allowed for a clear separation of the credit data into its fundamental components: a long-term trend, a repeating seasonal pattern, and a residual component. Comparing additive versus multiplicative models indicated which approach (constant vs. proportional seasonal effect) better explained the observed credit fluctuations based on residual variance.
- **Bank-Specific Trends and Currency Impact:** The individual bank plots revealed heterogeneous credit behaviors among institutions. The Coefficient of Variation (CV) analysis further quantified this variability, demonstrating which banks exhibit higher relative volatility in their loan portfolios. The analysis by currency (MN vs. ME) showed distinct distribution patterns and potentially different volatilities, suggesting that currency type is an important factor in understanding credit dynamics.
- **OLS Regression Insights:** The Ordinary Least Squares (OLS) regression indicated the statistical significance of 'Fecha' (time) and 'Moneda' (currency) in influencing credit amounts. The comparison of R-squared values provided a quantitative measure of how much additional explanatory power is gained by including currency as a predictor, supporting its relevance.
- **Stationarity and Autocorrelation Analysis:** The Augmented Dickey-Fuller (ADF) test revealed that the raw credit series were typically non-stationary, necessitating transformations like differencing and log-transformation. The ACF and PACF plots on the stationary series were crucial for identifying the autoregressive (AR) and moving average (MA) components, which are vital for specifying the orders of ARIMA and SARIMA models. The Ljung-Box test on model residuals confirmed whether remaining patterns (autocorrelation) were adequately captured.
- **SARIMA and ARIMA Model Performance:** Both SARIMA and ARIMA models provided forecasts, with SARIMA being particularly relevant given the clear seasonality identified in the data. Evaluation metrics (RMSE, MAPE) quantified the models' predictive accuracy. The model summaries offered insights into the statistical significance of various parameters. While the models provided valuable forecasts, the presence of heteroscedasticity (non-constant variance of errors, typically seen in residual plots or diagnostic tests in a full summary) indicates that some volatility patterns might not be fully captured, suggesting avenues for more advanced modeling.

A reference to Table **??**.

VII. Conclusions and Last Thoughts

The ability to accurately predict the demand and volume of agricultural loans is fundamental for the strategic management of financial institutions. Our study has culminated in the development of a SARIMA(1,1,1)(1,1,0,12) model that, by analyzing the intrinsic temporal structure of the data, offers forecasts with remarkable precision, evidenced by the low MAPE and RMSE values. The SARIMA model, by explicitly considering the inherent seasonality in agricultural credit flows (linked to planting, harvesting cycles, and annual climatic events), becomes an invaluable tool for Artificial Intelligence and decision support systems. The generated forecasts can directly feed into resource planning modules, liquidity optimization, and risk assessment. An AI system could, for example, use these projections to recommend capital allocations, adjust loan portfolios, or even identify periods of higher or lower credit demand. While the SARIMA model has proven effective in capturing the internal dynamics of the time series, it is imperative to recognize that loans to the agricultural sector are intrinsically linked to a complex network of external factors not captured solely by the historical behavior of the series. Factors such as climatic conditions (droughts, floods), inflation trends, fluctuations in exchange rate variations in the agricultural export market, and changes in monetary policy (interest rates) directly impact production capacity, farmers' income, and consequently, their credit demand and repayment capacity. We believe that the application of this model in an AI or decision-making environment is highly promising, provided it is complemented by a study and integration of the aforementioned external factors. To maximize predictive and prescriptive value, the next logical step would be to migrate towards more complex models such as SARIMAX (SARIMA with exogenous variables) or even explore Machine Learning algorithms (e.g., Random Forests, Gradient Boosting) that can model the non-linear relationships between credits and these macroeconomic or sectoral variables. The integration of these advanced forecasts will be of great help in projecting loans for the agricultural sector. It will allow banks to: Design and launch special campaigns dedicated to this sector at the most opportune moments, optimizing customer acquisition and product placement. Anticipate possible risk scenarios associated with the volatility of external factors and proactively adapt their credit policies.

In summary, our SARIMA model lays a solid foundation for understanding and predicting agricultural credit behavior. Its true potential will be unlocked when used as a key component within broader AI systems that incorporate and analyze the multifactorial influence of the environment, transforming forecasts into a tangible competitive advantage for banking decision-making.

REFERENCES

[1] "Time Series Forecasting" *Medium*, https://medium.com/@pooja_writes/time-series-forecasting-60652e12990c. Accessed on: May. 17, 2025.

[2] "Agricultural Loans Granted by Banks in Paraguay" *Banco Central del Paraguay*, https://www.bcp.gov.py. Accessed on: May. 13, 2025.